Unit-Root Testing

Econometrics II

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Outline: Unit-Root Testing

1. Properties of autoregressive unit-root processes.
2. Dickey-Fuller unit-root test.
   - AR(1).
   - AR(p).
   - Deterministic terms.
3. Further issues.

Properties of a Stationary AR(1)

- Consider the AR(1) model

$$y_t = \theta y_{t-1} + \epsilon_t, \quad t = 1, 2, ..., T.$$

The characteristic polynomial is $\theta(z) = 1 - \theta z$, with characteristic root $z_1 = \theta^{-1}$ and inverse root $\phi_1 = z_1^{-1} = \theta$. The stationarity condition is that $|\phi_1| = |\theta| < 1$.

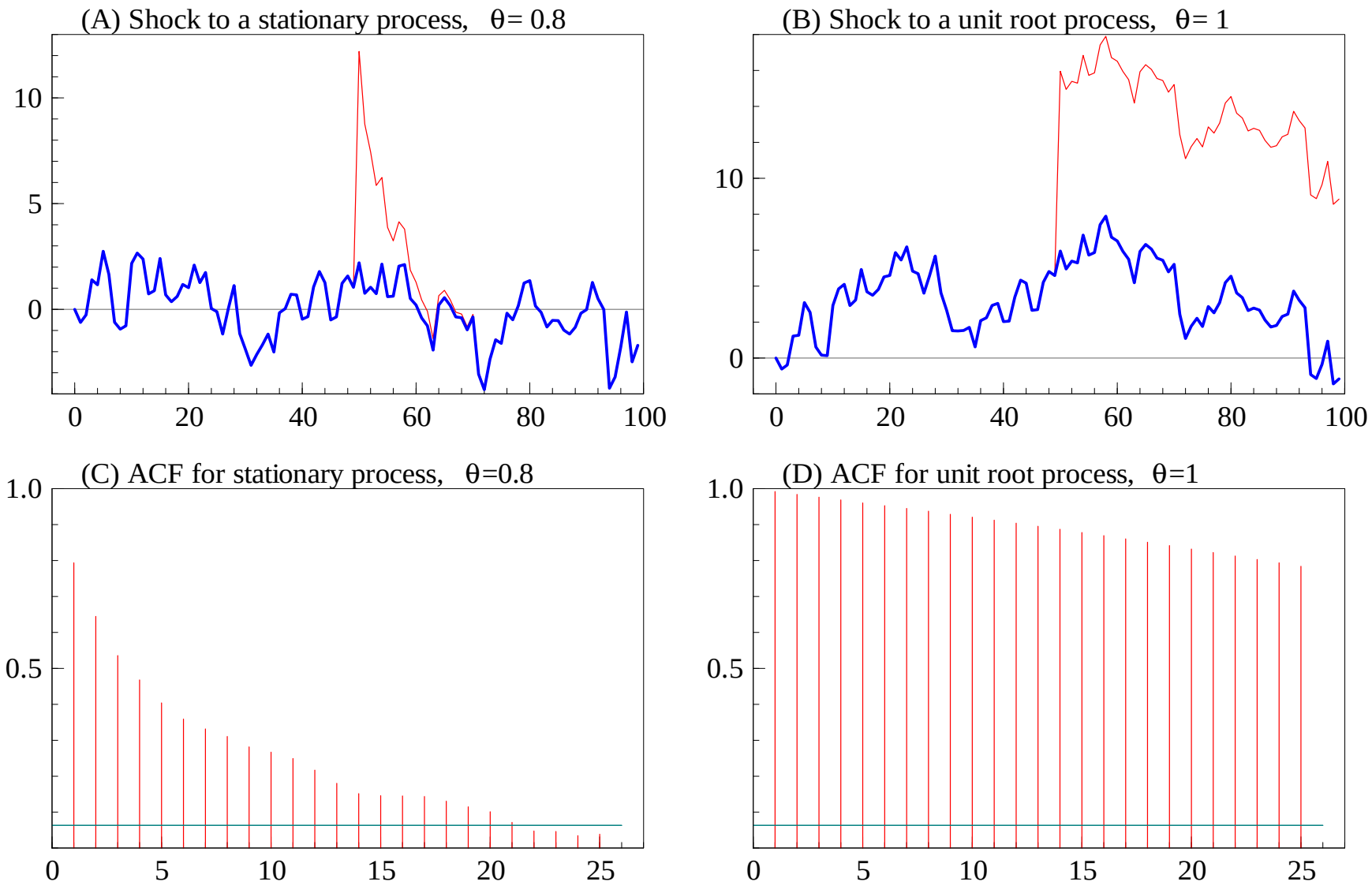
- Recall the solution

$$y_t = \epsilon_t + \theta\epsilon_{t-1} + \theta^2\epsilon_{t-2} + ... + \theta^{t-1}\epsilon_1 + \theta^t y_0,$$

where $\theta^s \to 0$. Shocks have only transitory effects. We have

$$\begin{aligned} E(y_t) &= \theta^t y_0 \to 0 \\ V(y_t) &= \sigma^2 + \theta^2\sigma^2 + \theta^4\sigma^2 + ... + \theta^{t-1}\sigma \to \frac{\sigma^2}{1-\theta^2} \\ \rho_s &= \mathrm{corr}(y_t, y_{t-s}) = \theta^s \end{aligned}$$

Simulated Example



Autoregression with a Unit Root

- Consider the AR(1) with $\theta = 1$, i.e.

$$y_t = y_{t-1} + \epsilon_t.$$

The characteristic polynomial is $\theta(z) = 1 - z$.
There is a unit root, $\theta(1) = 0$.

- The solution is given by

$$\begin{aligned} y_t &= y_0 + \Delta y_1 + \Delta y_2 + \ldots + \Delta y_t \\ &= y_0 + \epsilon_1 + \epsilon_2 + \ldots + \epsilon_t \\ &= y_0 + \sum_{i=1}^{t} \epsilon_i. \end{aligned}$$

Properties for the unit-root case, $\theta = 1$

1. The effect of y_0 stays in the process, and

$$E(y_t \mid y_0) = y_0.$$

2. The shock, ϵ_t, has permanent effects.
   It accumulates to a random walk component

$$\sum_{i=1}^{t} \epsilon_i,$$

   called a stochastic trend.

3. The variance is increasing,

$$V(y_t \mid y_0) = V\left(\sum_{i=1}^{t} \epsilon_i \mid y_0\right) = t \cdot \sigma^2,$$

   and the process y_t is clearly non-stationary.

4. The covariance, $\text{cov}(y_t, y_{t-s} \mid y_0)$, is given by

$$\begin{aligned} & E((y_t - y_0)(y_{t-s} - y_0) \mid y_0) \\ = \quad & E((\epsilon_1 + ... + \epsilon_t)(\epsilon_1 + ... + \epsilon_{t-s}) \mid y_0) = (t-s)\sigma^2. \end{aligned}$$

The autocorrelation is

$$\begin{aligned} \text{corr}(y_t, y_{t-s} \mid y_0) &= \frac{\text{cov}(y_t, y_{t-s} \mid y_0)}{\sqrt{V(y_t \mid y_0) \cdot V(y_{t-s} \mid y_0)}} \\ &= \frac{(t-s)\sigma^2}{\sqrt{t\sigma^2 \cdot (t-s)\sigma^2}} \\ &= \sqrt{\frac{t-s}{t}}, \end{aligned}$$

which dies out very slowly with s.

5. The first difference, $\Delta y_t = \epsilon_t$, is stationary.
y_t is called integrated of first order, $I(1)$.

Higher Order Processes

- For an AR(p) process, with $p > 1$, the MA-solution is more complicated.
- An I(1) process can be written as

$$y_t = c \sum_{i=1}^{t} \epsilon_i + c_0^* \epsilon_t + c_1^* \epsilon_{t-1} + c_2^* \epsilon_{t-2} + ... + c_{t-1}^* \epsilon_1 + A, \qquad (1)$$

where

1. A is the contribution from initial values.
2. The sequence $c_0^*, c_1^*, c_2^*, ...$ converges to zero exponentially fast.

Deterministic Terms

- To model actual time series we include deterministic terms, e.g.

$$y_t = \delta + \theta y_{t-1} + \epsilon_t.$$

With a unit root, the terms accumulate!

- If $|\theta| < 1$, the solution is

$$y_t = \theta^t y_0 + \sum_{i=0}^{t-1} \theta^i \epsilon_{t-i} + (1 + \theta + \theta^2 + ... + \theta^{t-1})\delta,$$

where the mean converges to $(1 + \theta + \theta^2 + ... + \theta^{t-1})\delta \to \delta/(1-\theta)$.

- If $\theta = 1$, the solution is

$$y_t = y_0 + \sum_{i=1}^{t}(\delta + \epsilon_i) = y_0 + \delta t + \sum_{i=1}^{t} \epsilon_i.$$

The constant term produces a deterministic linear trend.
y_t is called a random walk with drift.
Note the parallel between a deterministic and a stochastic trend.

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Unit Root Testing

- Estimate an autoregressive model and test whether

$$\theta(1) = 0,$$

i.e. whether $z = 1$ is a root in the autoregressive polynomial.

- This is a straightforward hypothesis test!

1. Careful statistical model
   Which deterministic components are relevant: constant or trend?
2. Hypothesis
   What are the properties of the model under H_0 and H_A.
   Are both models relevant?
3. Test statistic
   What is a relevant test statistic for $H_0 \mid H_A$?
4. Asymptotic distribution
   The asymptotic distribution for a unit root test is non-standard.

Dickey-Fuller Test in an AR(1)

- Consider an AR(1) model

$$y_t = \theta y_{t-1} + \epsilon_t, \quad \text{with } \epsilon_t \text{ i.i.d.}(0, \sigma_0^2).$$

The *one-sided* hypothesis is $\theta(1) = 1 - \theta = 0$, against stationarity:

$$H_0 : \theta = 1 \quad \text{against} \quad H_A : -1 < \theta < 1.$$

An equivalent formulation is

$$\Delta y_t = \pi y_{t-1} + \epsilon_t,$$

where $\pi = \theta - 1 = -\theta(1)$. The hypothesis $\theta(1) = 0$ translates into

$$H_0 : \pi = 0 \quad \text{against} \quad H_A : -2 < \pi < 0.$$

- The Dickey-Fuller (DF) test statistic is simply the $t-$ ratio, i.e.

$$\hat{\tau} = \frac{\hat{\theta} - 1}{\text{se}(\hat{\theta})} = \frac{\hat{\pi}}{\text{se}(\hat{\pi})}.$$

The asymptotic distribution is Dickey-Fuller, DF, and not $N(0, 1)$.

Asymptotic Analysis

- The likelihood function for the AR(1) model is

$$\log L_T(\theta) = \sum_{t=1}^{T} \ell_t(\theta) \quad \text{with} \quad \ell_t(\theta) = -\frac{1}{2}\log\left(2\pi\sigma_0^2\right) - \frac{\left(y_t - \theta y_{t-1}\right)^2}{2\sigma_0^2}.$$

The score contributions is therefore the first derivatives

$$s_t\left(\theta\right) = \frac{\partial \log \ell_t(\theta)}{\partial \theta} = \frac{y_{t-1}\left(y_t - \theta y_{t-1}\right)}{\sigma_0^2},$$

and the MLE is

$$\hat{\theta} = \frac{\sum_{t=1}^{T} y_{t-1} y_t}{\sum_{t=1}^{T} y_{t-1}^2} = \theta_0 + \frac{\frac{1}{T}\sum_{t=1}^{T} y_{t-1}\epsilon_t}{\frac{1}{T}\sum_{t=1}^{T} y_{t-1}^2}.$$

- The second derivative of the log-likelihood contribution is given by

$$\frac{\partial^2 \log \ell_t(\theta)}{\partial\theta\partial\theta} = \frac{\partial}{\partial\theta}\left\{\frac{y_{t-1}\left(y_t - \theta y_{t-1}\right)}{\sigma_0^2}\right\} = -\frac{y_{t-1}^2}{\sigma_0^2}.$$

- In the unit root case, $y_t = \sum_{i=1}^t \epsilon_i$, and the average

$$\frac{1}{T}\sum_{t=1}^{T} y_{t-1}^2$$

does *not* converge to a constant, because $V(y_t \mid y_0)$ grows with t.

- Instead it hold that

$$\frac{1}{T^2}\sum_{t=1}^{T} y_{t-1}^2 \xrightarrow{d} \sigma_0^2 \int_0^1 B(u)^2 du,$$

where $B(u)$ is a so-called Brownian motion defined on $u \in [0, 1]$.

- This result is different from the usual law of large numbers:
  1. The denominator is T^2 and not T. Information on $\hat{\theta}$ grows faster.
  2. The right hand side limit is stochastic and not a constant.

- The normalized score converges to a stochastic intergral,

$$\frac{1}{T}\sum_{t=1}^{T} y_{t-1}\epsilon_t \xrightarrow{d} \sigma_0^2 \int_0^1 B(u)dB(u).$$

Jointly, the estimator has a complicated asymptotic distribution

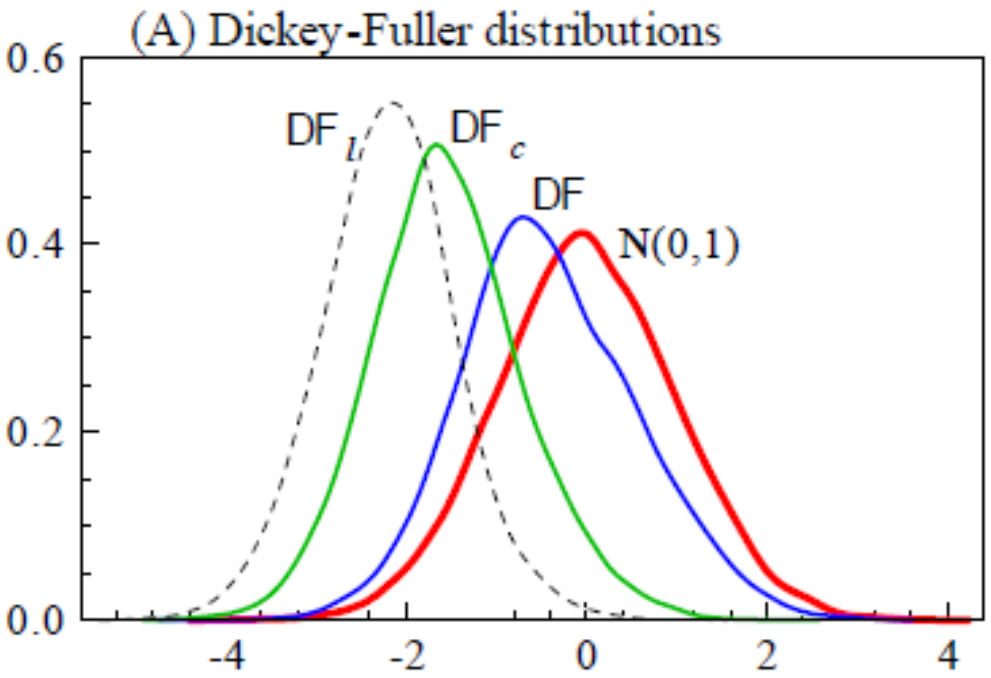
$$T(\hat{\theta}-1) \xrightarrow{d} \frac{\int_0^1 B(u)dB(u)}{\int_0^1 B(u)^2 du}.$$

- Note that the speed of convergence is T instead of the usual $\sqrt{T}$. We say that the estimator is super-consistent.

- Likewise the DF distribution can be written as

$$\hat{\tau} \xrightarrow{d} \frac{\int_0^1 B(u)dB(u)}{\sqrt{\int_0^1 B(u)^2 du}}.$$

Distribution	Quantile			
	1%	2.5%	5%	10%
$N(0,1)$	-2.33	-1.96	-1.64	-1.28
DF	-2.56	-2.23	-1.94	-1.62
DF_c	-3.43	-3.12	-2.86	-2.57
DF_l	-3.96	-3.66	-3.41	-3.13



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Dickey-Fuller Test in an AR(p)

- The DF distribution is derived under the assumption that ϵ_t is IID. For an AR(p), we use the Augmented Dickey-Fuller (ADF) test.

- Consider the case of $p = 3$ lags:

$$y_t = \theta_1 y_{t-1} + \theta_2 y_{t-2} + \theta_3 y_{t-3} + \epsilon_t.$$

A unit root in $\theta(z) = 1 - \theta_1 z - \theta_2 z^2 - \theta_3 z^3$ corresponds to

$$\theta(1) = 1 - \theta_1 - \theta_2 - \theta_3 = 0.$$

- This hypothesis is straightforward to test using

$$\hat{\tau} = \frac{\sum_{i=1}^{3} \hat{\theta}_i - 1}{\text{se}\left(\sum_{i=1}^{3} \hat{\theta}_i\right)},$$

provided that we have the sum, $\sum_{i=1}^{3} \hat{\theta}_i$, and the standard error.

- To avoid testing on the sum, $\sum_{i=1}^{3} \hat{\theta}_i$, the model is rewritten as

$$
\begin{aligned}
y_t - y_{t-1} &= (\theta_1 - 1)y_{t-1} + \theta_2 y_{t-2} + \theta_3 y_{t-3} + \epsilon_t \\
y_t - y_{t-1} &= (\theta_1 - 1)y_{t-1} + (\theta_2 + \theta_3)y_{t-2} + \theta_3(y_{t-3} - y_{t-2}) + \epsilon_t \\
y_t - y_{t-1} &= (\theta_1 + \theta_2 + \theta_3 - 1)y_{t-1} - (\theta_2 + \theta_3)\Delta y_{t-1} - \theta_3 \Delta y_{t-2} + \epsilon \\
\Delta y_t &= \pi y_{t-1} + c_1 \Delta y_{t-1} + c_2 \Delta y_{t-2} + \epsilon_t,
\end{aligned}
$$

where

$$
\pi = \theta_1 + \theta_2 + \theta_3 - 1 = -\theta(1), \quad c_1 = -\left(\theta_2 + \theta_3\right), \quad c_2 = -\theta_3.
$$

- Now the ADF test is simply

$$
\hat{\tau} = \frac{\sum_{i=1}^{3} \hat{\theta}_i - 1}{\mathrm{se}(\sum_{i=1}^{3} \hat{\theta}_i)} = \frac{\hat{\pi}}{\mathrm{se}(\hat{\pi})},
$$

similar to the AR(1) case.

Remarks:

1. It is only the test for $\pi = 0$ that follows the DF distribution. Tests on c_1 and c_2 are $N(0, 1)$.

2. We use the normal tools to determine the appropriate lag-length: general-to-specific testing or information criteria.

3. Verbeek suggests to calculate the DF test for all values of p. ... but why should we look at inferior or misspecified models? Often better to find the best model and test in that.

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Dickey-Fuller Test with a Constant Term

- We need deterministic variables to allow $E(y_t) \neq 0$ under H_A.
  The DF regression with a constant term (and $p = 3$ lags again) is

$$\Delta y_t = \delta + \pi y_{t-1} + c_1 \Delta y_{t-1} + c_2 \Delta y_{t-2} + \epsilon_t. \qquad (*)$$

  The hypothesis is unchanged $H_0 : \pi = 0$, and the test statistic is

$$\hat{\tau}_c = \frac{\hat{\pi}}{\text{se}(\hat{\pi})}.$$

- Remarks:
  1. The constant term changes the asymptotic distribution.
     The relevant distribution, DF_c, is shifted to the left of DF.
  2. For an AR(1) process, under the null hypothesis, $\pi = 0$, the constant gives a trend in y_t:

$$y_t = \begin{cases} \mu & + \quad \text{stationary process} & & \text{for} \quad |\theta| < 1 \\ y_0 & + \quad \text{random walk} & + \quad \delta t & \text{for} \quad \theta = 1 \end{cases}$$

     That is not a natural comparison. We assume that $\delta = 0$ if $\theta = 1$.

- A more satisfactory hypothesis is $H_0^* : \pi = \delta = 0$. Compare $(*)$ with

$$\Delta y_t = c_1 \Delta y_{t-1} + c_2 \Delta y_{t-2} + \epsilon_t. \qquad (**)$$

- The joint hypothesis can be tested by a LR test,

$$LR(\pi = \delta = 0) = -2 \cdot (\log L_0 - \log L_A),$$

where $\log L_0$ and $\log L_A$ denote the log-likelihood from $(*)$ and $(**)$.

- $LR(\pi = \delta = 0)$ follows a non-standard distribution, DF_c^2.

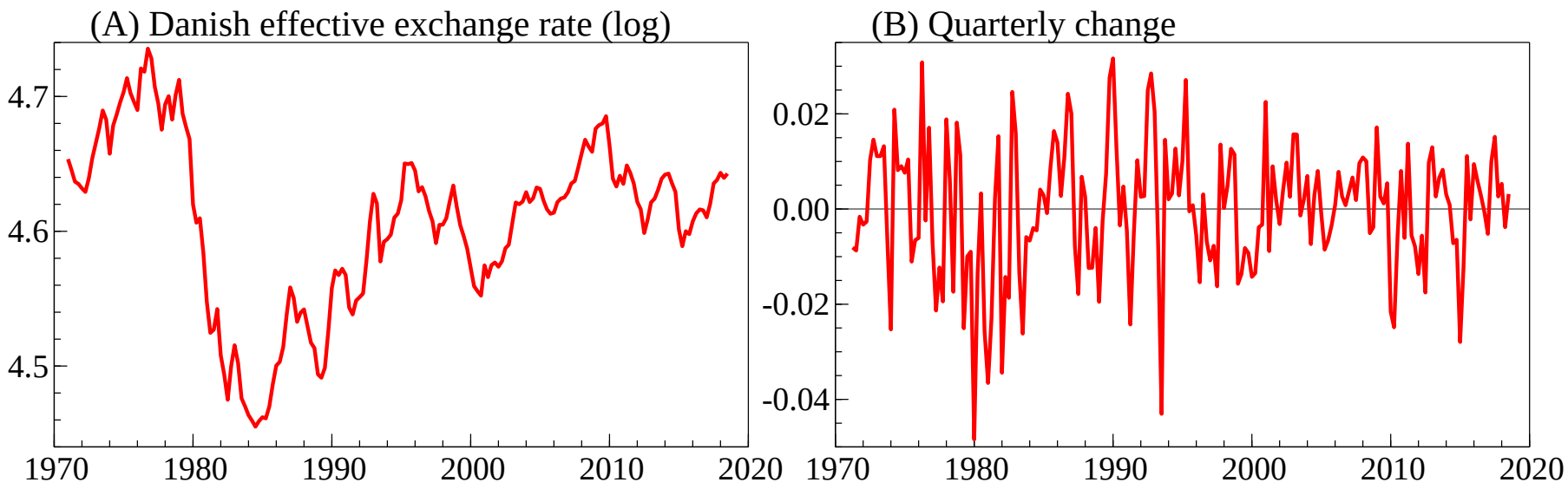
	Nominal Level			
Distribution	1%	2.5%	5%	10%
$\chi^2(2)$	9.21	7.38	5.99	4.61
DF_c^2	12.73	10.73	9.13	7.50
DF_l^2	16.39	14.13	12.39	10.56

Empirical Example: Danish Effective Exchange Rate

Consider quarterly data from 1971:1 for the log of nominal effective exchange rate,

$$e_t = \log(\text{EFKRKS}_t).$$

We allow a constant in the model.



- An AR(2) model for 1971:3-2018:3 produces

	Coefficient	Std.Error	t-value
De_1	0.311025	0.06959	4.47
Constant	0.137878	0.06865	2.01
e_1	-0.0299348	0.01490	-2.01
sigma	0.0129549		
R^2	0.107432		
log-likelihood	554.78		
no. of observations	189		

- The DF $t-$test statistic is

$$\hat{\tau}_c = \frac{-0.02993}{0.01490} = -2.01.$$

The critical value in the DF_c distribution is -2.86.
We do not reject a unit root.

- To test the joint hypothesis, $H_0^* : \pi = \delta = 0$, we estimate the model under the null hypothesis:

	Coefficient	Std.Error	t-value
De_1	0.296541	0.06959	4.26
sigma	0.0130247		
log-likelihood	552.753		
no. of observations	189		

- The LR test statistic is given by

$$LR(\pi = \delta = 0) = 2 \cdot (554.780 - 552.753) = 4.054.$$

The critical value in the DF_c^2 distribution is 9.13.
We do not reject a unit root.

Dickey-Fuller Test with a Trend Term

- For trending variables, the relevant alternative is trend-stationarity. We use

$$\Delta y_t = \delta + \gamma t + \pi y_{t-1} + c_1 \Delta y_{t-1} + c_2 \Delta y_{t-2} + \epsilon_t. \qquad (\#)$$

  The hypothesis is still $H_0 : \pi = 0$, and the DF $t-$test statistic is

$$\hat{\tau}_l = \frac{\hat{\pi}}{\text{se}(\hat{\pi})}.$$

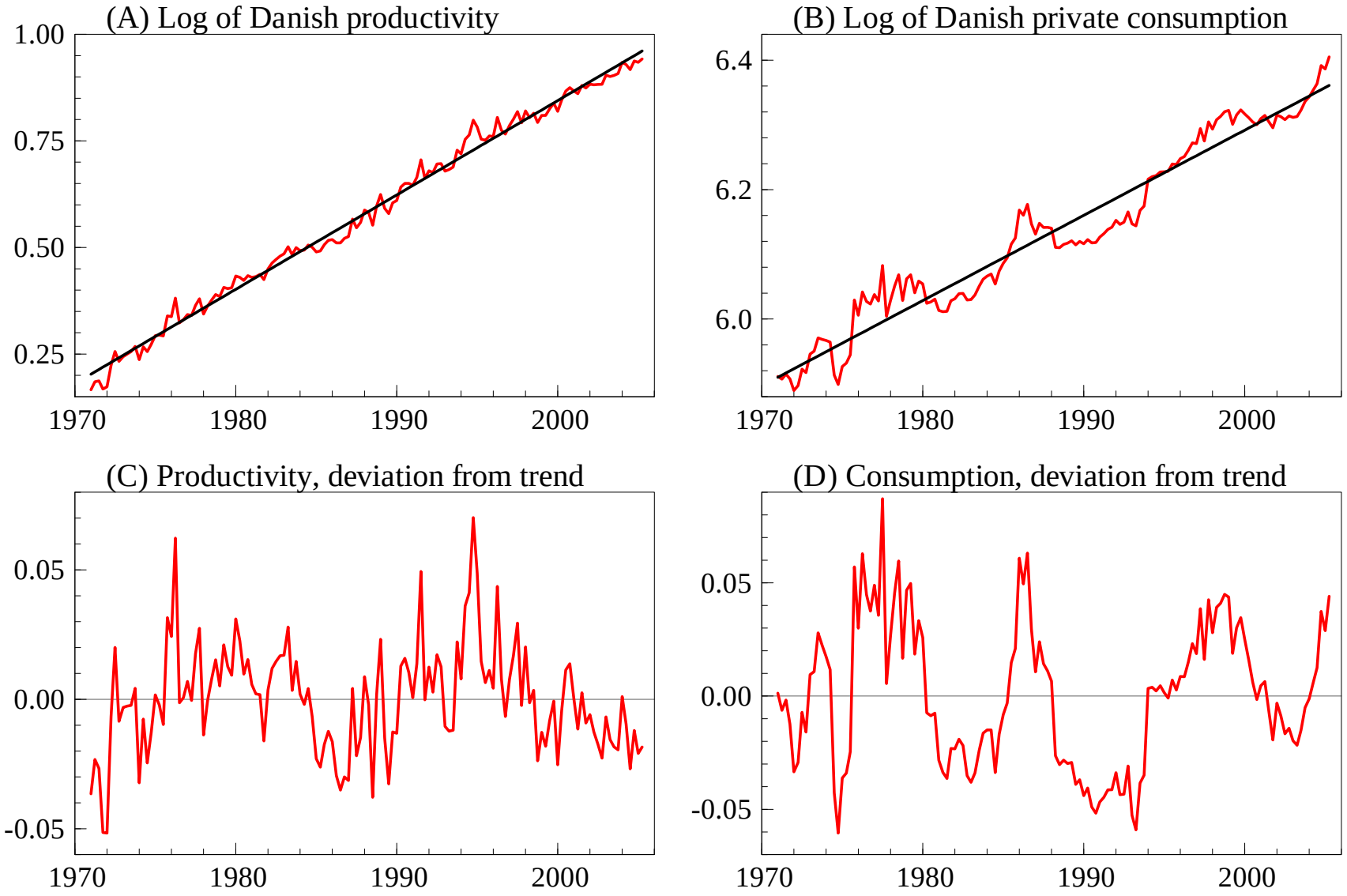
  The presence of a trend shifts the asymptotic distribution, DF_l, further to the left.

- To avoid the accumulation of the trend under $\pi = 0$, we may consider the joint hypothesis, $H_0^* : \pi = \gamma = 0$, i.e. to compare $(\#)$ with

$$\Delta y_t = \delta + c_1 \Delta y_{t-1} + c_2 \Delta y_{t-2} + \epsilon_t. \qquad (\#\#)$$

  The LR test statistic is $LR(\pi = \gamma = 0) = -2 \cdot (\log L_0 - \log L_A)$, which follows a DF_l^2.

Empirical Examples: Trend-Stationarity



Empirical Example: Danish Productivity

- To test if log-productivity is trend-stationary we use

$$\Delta \text{LPROD}_t = \underset{(-6.22)}{-0.439}\text{LPROD}_{t-1} + \underset{(6.58)}{0.091} + \underset{(6.15)}{0.0024}t + \hat{\epsilon}_t,$$

with $\log L_A = 366.09$. The DF $t-$test statistic is $\hat{\tau}_l = -6.22 \ll -3.41$.
We reject and conclude that productivity is trend-stationary.

- To test the joint hypothesis, $H_0^* : \pi = \gamma = 0$, we use

$$\Delta \text{LPROD}_t = \underset{(3.48)}{0.0057} + \epsilon_t,$$

with $\log L_0 = 348.63$. The LR test statistic is

$$LR(\pi = \gamma = 0) = -2 \cdot (348.63 - 366.09) = 34.92.$$

This is larger than the critical value of 12.39 in DF_l^2, and we reject the unit root hypothesis.

Empirical Example: Danish Consumption

- To test if log-consumption is trend-stationary we use the regression

$$\Delta \mathrm{LC}_t = \underset{(-2.56)}{-0.129}\mathrm{LC}_{t-1} \underset{(-2.43)}{-0.209}\,\Delta \mathrm{LC}_{t-1} + \underset{(2.57)}{0.764} + \underset{(2.58)}{0.0004}t + \hat{\epsilon}_t,$$

  with $\log L_A = 359.23$. The Dickey-Fuller $t-$test statistic is $\hat{\tau}_l = -2.56$, which is not significant in the DF_l distribution. We conclude that private consumption seems to have a unit-root.

- To test the joint hypothesis, $H_0^* : \pi = \gamma = 0$, we use

$$\Delta \mathrm{LC}_t = \underset{(-3.29)}{-0.274}\Delta \mathrm{LC}_{t-1} + \underset{(2.97)}{0.0046} + \epsilon_t,$$

  with $\log L_0 = 355.87$. The LR test statistic for a unit root is

$$LR(\pi = \gamma = 0) = -2 \cdot (355.87 - 359.23) = 6.72,$$

  which is smaller than 12.39. We cannot reject a unit root.

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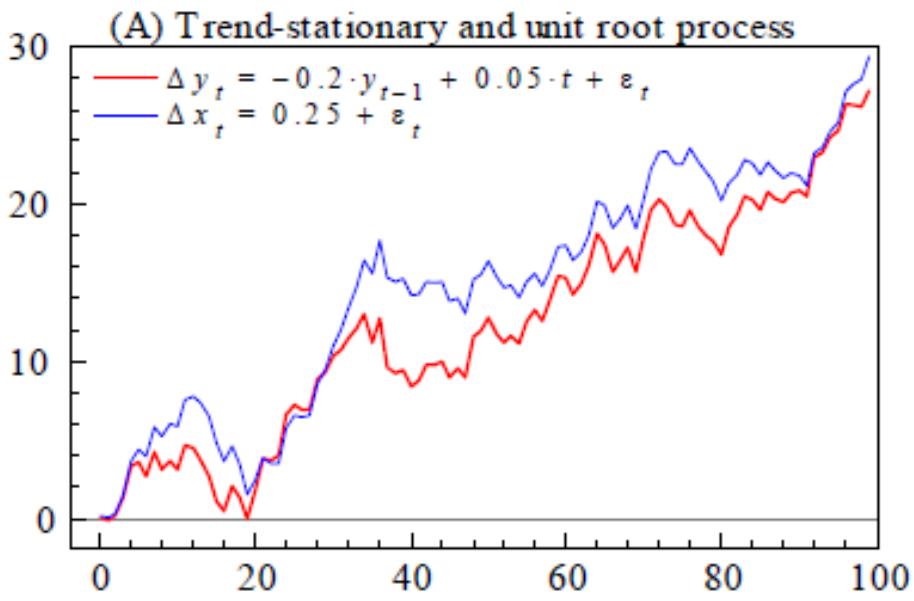
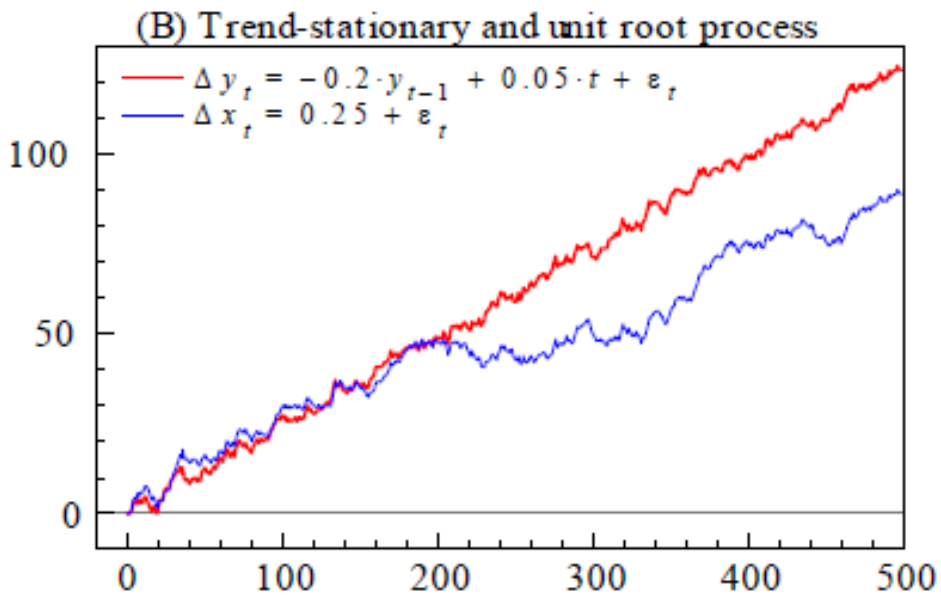
The Problem of Low Power

- It is difficult to distinguish unit roots from large stationary roots. Always be careful in conclusions.
- Consider time series generated from the two models

$$\begin{aligned} \Delta y_t &= -0.2 \cdot y_{t-1} + 0.05 \cdot t + \epsilon_t \\ \Delta x_t &= 0.25 + \epsilon_t. \end{aligned}$$

Hard to tell apart in practice. We need many observations to be sure.





Special Events

- Unit root tests assess whether shocks are transitory or permanent.
  The conclusions are sensitive to a few large shocks.

- Consider a one-time change in the mean of the series, a break.
  This is one large shock with a permanent effect.
  Even if the series is piece-wise stationary, such that normal shocks have transitory effects, the presence of a break will make it look like the shocks have permanent effects.
  That may bias the conclusion towards a unit root.

- Consider a few large outliers.
  The series may look more mean reverting than it actually is.
  That may bias the results towards stationarity.